

HOMEWORK 2: SOLUTIONS

1. (Fermat's little theorem) Fermat's little theorem asserts that for any prime p and any $a \in \mathbb{Z}$, we have $a^p \equiv a \pmod{p}$. In class we deduced this result from Euler's theorem. Here we give a direct proof.

(1) Consider the binomial coefficient

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}, \quad 1 \leq k \leq p-1,$$

which are all positive integers. Prove that p divides $\binom{p}{k}$ for every $1 \leq k \leq p-1$.

Solution. Since $p \mid p!$ and

$$\binom{p}{k} k!(p-k)! = p!,$$

p divides the left-hand side. For $1 \leq k < p$, we have $1 \leq p-k < p$. Thus, $p \nmid k!$ and $p \nmid (p-k)!$. By Euclid's lemma, $p \mid \binom{p}{k}$.

(2) Show that for any $x, y \in \mathbb{Z}$,

$$(x+y)^p \equiv x^p + y^p \pmod{p}.$$

Then use this and induction to show that for any $x_1, \dots, x_n \in \mathbb{Z}$,

$$\left(\sum_{i=1}^n x_i \right)^p \equiv \sum_{i=1}^n x_i^p \pmod{p}. \quad (1)$$

Solution. By the binomial theorem,

$$(x+y)^p = x^p + \sum_{k=1}^{p-1} \binom{p}{k} x^{p-k} y^k + y^p.$$

Since each binomial coefficient in the sum is divisible by p , we have

$$(x+y)^p \equiv x^p + y^p \pmod{p}.$$

The general congruence (1) holds trivially for $n = 1$. Our argument above shows that it also holds for $n = 2$. Suppose now that it holds for some positive integer n and any $x_1, \dots, x_n \in \mathbb{Z}$. Using this and the case $n = 2$, we obtain

$$\left(\sum_{i=1}^{n+1} x_i \right)^p = \left(\sum_{i=1}^n x_i + x_{n+1} \right)^p \equiv \left(\sum_{i=1}^n x_i \right)^p + x_{n+1}^p \equiv \sum_{i=1}^n x_i^p + x_{n+1}^p \equiv \sum_{i=1}^{n+1} x_i^p \pmod{p}$$

for any $x_1, \dots, x_{n+1} \in \mathbb{Z}$. This finishes the inductive proof of (1).

(3) Deduce Fermat's little theorem.

Solution. The theorem is obviously true when $a = 0$. It is also true when $p = 2$, since a^2 and a have the same parity.

Suppose now that $a \neq 0$ and $p > 2$ is odd. Let

$$\epsilon_a = \begin{cases} 1, & \text{if } a > 0, \\ -1, & \text{if } a < 0, \end{cases}$$

so that $a = \epsilon_a |a|$. We apply (1) with $n = |a|$ and $x_1 = \cdots = x_n = 1$ to get $|a|^p \equiv |a| \pmod{p}$. Since $\epsilon_a^p = \epsilon_a$, we have

$$a^p = (\epsilon_a |a|)^p = \epsilon_a^p |a|^p = \epsilon_a |a|^p \equiv \epsilon_a |a| \equiv a \pmod{p}.$$

2. (Euler's theorem) Let $\phi(m)$ be the Euler totient function, which, for every positive integer m , counts the number of positive integers $n \leq m$ with $\gcd(n, m) = 1$. In class we proved Euler's theorem that if $\gcd(a, m) = 1$, then $a^{\phi(m)} \equiv 1 \pmod{m}$.

- (1) Show that if p^k is a prime power, then $\phi(p^k) = p^{k-1}(p-1)$. This is a natural generalization of $\phi(p) = p-1$ discussed in class.

Solution. Consider the set $A = \{1, 2, \dots, p^k\}$. Observe that $\gcd(n, p^k) = 1$ if and only if $p \nmid n$. For each $n \in A$ divisible by p , write $n = pm$ for some positive integer m . Since $0 < n \leq p^k$, we have $0 < m \leq p^{k-1}$. So the number of choices for m is p^{k-1} . In other words, the number of $n \in A$ divisible by p is p^{k-1} . Hence

$$\phi(p^k) = |A| - p^{k-1} = p^k - p^{k-1} = p^{k-1}(p-1).$$

- (2) Compute the following residues. For each part, your final answer should be a nonnegative integer less than the given modulus.

- (a) $18^{20} \pmod{25}$

Solution. Since $\gcd(18, 25) = 1$ and $\phi(25) = 5(5-1) = 20$, Euler's theorem implies $18^{20} \equiv 1 \pmod{25}$.

- (b) $3^{2026} \pmod{49}$

Solution. Since $\gcd(3, 49) = 1$ and $\phi(49) = 7(7-1) = 42$, Euler's theorem implies $3^{42} \equiv 1 \pmod{49}$. Writing $2026 = 42 \cdot 48 + 10$, we have

$$3^{2026} = (3^{42})^{48} \cdot 3^{10} \equiv 3^{10} \pmod{49}.$$

According to the hint,

$$3^5 = 3^4 \cdot 3 = 81 \cdot 3 \equiv 32 \cdot 3 \equiv 96 \equiv -2 \pmod{49}.$$

So

$$3^{2026} \equiv 3^{10} \equiv (3^5)^2 \equiv (-2)^2 \equiv 4 \pmod{49}.$$

- (c) $4^{137893} \pmod{121}$

Solution. First observe that $4^{137}8^{93} = 2^{553}$. Since $\gcd(2, 121) = 1$ and $\phi(121) = 11(11 - 1) = 110$, Euler's theorem implies $2^{110} \equiv 1 \pmod{121}$. Since $553 = 110 \cdot 5 + 3$, we have

$$4^{137}8^{93} = 2^{553} = (2^{110})^5 \cdot 2^3 \equiv 2^3 \equiv 8 \pmod{121}.$$

(3) Euler's theorem can also be stated in the form

$$\bar{a} \equiv a^{\phi(m)-1} \pmod{m},$$

where $\bar{a}$ is the least positive multiplicative inverse of $a \pmod{m}$, meaning that it is the least positive integer satisfying $a\bar{a} \equiv 1 \pmod{m}$. Use this to find the least positive multiplicative inverse of $3 \pmod{25}$.

Solution. Using $\phi(25) = 20$ from (2a), we get

$$\bar{3} \equiv 3^{20-1} \equiv 3^{19} \pmod{25}.$$

Because $19 = 3 \cdot 6 + 2$ and $3^3 = 27 \equiv 2 \pmod{25}$, we have

$$\bar{3} \equiv 3^{19} = (3^3)^6 \cdot 3 \equiv 2^6 \cdot 3 \equiv 64 \cdot 3 \equiv 14 \cdot 3 \equiv 42 \equiv 17 \pmod{25}.$$

So the least positive multiplicative inverse of $3 \pmod{25}$ is 17.

(4) For an odd positive integer m with prime factorization $m = p_1^{n_1} \cdots p_k^{n_k}$, where $p_1, \dots, p_k$ are distinct primes and $n_1, \dots, n_k$ are positive integers, the *Carmichael function* $\lambda(m)$ is defined as the least common multiple of $\phi(p_1^{n_1}), \dots, \phi(p_k^{n_k})$, namely, the least positive integer divisible by each of these numbers, with the convention that $\lambda(1) = 1$. Assuming the property that $\phi(n) \mid \phi(m)$ whenever $n \mid m$, show that $\lambda(m) \mid \phi(m)$ and that if $\gcd(a, m) = 1$, then

$$a^{\lambda(m)} \equiv 1 \pmod{m}. \quad (2)$$

For comparison, when $m = 15$, we have $\phi(15) = 8$ and $\lambda(15) = 4$. So the above becomes $a^4 \equiv 1 \pmod{15}$, while Euler's theorem gives the weaker conclusion $a^8 \equiv 1 \pmod{15}$.

Solution. By the division algorithm, there are $q, r \in \mathbb{Z}$ such that

$$\phi(m) = \lambda(m)q + r, \quad 0 \leq r < \lambda(m). \quad (3)$$

For each prime power p^{n_i} , we have $\phi(p^{n_i}) \mid \phi(m)$ by assumption and $\phi(p^{n_i}) \mid \lambda(m)$ by definition. By (3), $\phi(p^{n_i}) \mid r$. Hence, r is a common multiple of $\phi(p_1^{n_1}), \dots, \phi(p_k^{n_k})$. Since $\lambda(m)$ is the least such common multiple and $0 \leq r < \lambda(m)$, we have $r = 0$. Hence, $\lambda(m) \mid \phi(m)$.

To prove (2), observe that Euler's theorem gives $a^{\phi(p^{n_i})} \equiv 1 \pmod{p^{n_i}}$. Since $\phi(p^{n_i}) \mid \lambda(m)$, we have

$$a^{\lambda(m)} = \left(a^{\phi(p^{n_i})}\right)^{\frac{\lambda(m)}{\phi(p^{n_i})}} \equiv 1 \pmod{p^{n_i}}.$$

Since $p_1^{n_1}, \dots, p_k^{n_k}$ are pairwise relatively prime, applying the hint successively gives

$$a^{\lambda(m)} \equiv 1 \pmod{p_1^{n_1}}, \quad a^{\lambda(m)} \equiv 1 \pmod{p_1^{n_1}p_2^{n_2}}, \quad \dots, \quad a^{\lambda(m)} \equiv 1 \pmod{\underbrace{p_1^{n_1}p_2^{n_2} \cdots p_k^{n_k}}_m}.$$

3. (Lagrange's theorem) Recall that $\mathbb{Z}[x]$ denotes the ring of polynomials with integer coefficients. For any nonzero $m \in \mathbb{Z}$, we may extend the congruence relation modulo m to $\mathbb{Z}[x]$ by defining $f(x) \equiv g(x) \pmod{m}$ if m divides every coefficient of $f(x) - g(x)$. It is easy to check that $\equiv$ is an equivalence relation on $\mathbb{Z}[x]$.

- (1) Let $a \in \mathbb{Z}$ and $f(x) \in \mathbb{Z}[x]$. Show that $f(a) \equiv 0 \pmod{m}$ if and only if there is some $g(x) \in \mathbb{Z}[x]$ such that

$$f(x) \equiv (x - a)g(x) \pmod{m}. \quad (4)$$

Solution. If (4) holds, we simply need to plug in $x = a$ to see $f(a) \equiv 0 \pmod{m}$.

Conversely, let us assume $f(a) \equiv 0 \pmod{m}$. Suppose that

$$f(x) = \sum_{k=0}^n c_k x^k \in \mathbb{Z}[x].$$

Then

$$f(x) - f(a) = \sum_{k=0}^n c_k x^k - \sum_{k=0}^n c_k a^k = \sum_{k=0}^n c_k (x^k - a^k).$$

According to the hint, for each $0 \leq k \leq n$, we have $x^k - a^k = (x - a)g_k(x)$ with

$$g_k(x) = x^{k-1} + ax^{k-2} + a^2x^{k-3} + \cdots + a^{k-2}x + a^{k-1} \in \mathbb{Z}[x].$$

Using $f(a) \equiv 0 \pmod{m}$, we see that

$$f(x) \equiv f(x) - f(a) \equiv \sum_{k=0}^n c_k (x^k - a^k) \equiv (x - a) \sum_{k=0}^n c_k g_k(x) \pmod{m}.$$

Define

$$g(x) = \sum_{k=0}^n c_k g_k(x).$$

Then $g(x) \in \mathbb{Z}[x]$ and (4) holds.

- (2) Prove Lagrange's theorem that if p is prime and

$$f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x],$$

where $n \geq 0$ and $p \nmid a_n$, then $f(x) \equiv 0 \pmod{p}$ has at most n roots.

Solution. The case $n = 0$ is clear. Assume that Lagrange's theorem holds for all polynomials in $\mathbb{Z}[x]$ of some common degree $n \geq 0$. Let

$$f(x) = a_{n+1} x^{n+1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$$

be a polynomial of degree $n + 1$ with $p \nmid a_{n+1}$. Part (1) shows that if a is a root of $f(x) \equiv 0 \pmod{p}$, then there is some $g(x) \in \mathbb{Z}[x]$ such that

$$f(x) \equiv (x - a)g(x) \pmod{p}.$$

Observe that $g(x)$ has the same leading coefficient a_{n+1} and degree n . By Euclid's lemma, every other root b of $f(x) \equiv 0 \pmod{p}$ must also be a root of $g(x) \equiv 0 \pmod{p}$. By inductive hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most n roots. Hence, $f(x) \equiv 0 \pmod{p}$ has at most $n + 1$ roots. This completes the inductive proof of Lagrange's theorem.

4. (Euler's criterion) Euler's criterion states that if $p > 2$ is prime and $a \in \mathbb{Z}$ with $p \nmid a$, then

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

We proved this in class based on Lagrange's theorem. The purpose of this problem is to obtain a different proof. Consider the set $A = \{1, \dots, p-1\}$.

- (1) Show that when $a \not\equiv 1 \pmod{p}$, A can be partitioned into $\frac{p-1}{2}$ pairs of distinct numbers, such that each pair $\{x, y\}$ satisfies $xy \equiv a \pmod{p}$, and that when $a \equiv 1 \pmod{p}$, A can be partitioned into $\frac{p-1}{2}$ pairs of distinct numbers, such that one pair is of the form $\{x_0, p-x_0\}$ satisfying $x_0^2 \equiv a \pmod{p}$ and every other pair $\{x, y\}$ satisfies $xy \equiv a \pmod{p}$.

Solution. The theorem on linear congruences implies that given any $x \in A$, which has $\gcd(x, p) = 1$, there is a unique element $y = f(x) \in A$ such that $xf(x) \equiv a \pmod{p}$. This defines a bijection from A onto itself.

When $a \not\equiv 1 \pmod{p}$, there is no $x \in A$ such that $f(x) = x$. So we may pair elements of A as $\{x, f(x)\}$ with the property that $xf(x) \equiv a \pmod{p}$, and there are exactly $\frac{p-1}{2}$ such pairs. This gives the described partition of A .

When $a \equiv 1 \pmod{p}$, the congruence $x^2 \equiv a \pmod{p}$ has exactly two solutions given by some $x_0 \in A$ and $p-x_0 \in A$, which are two distinct elements of A due to p being odd. We pair these two as $\{x_0, p-x_0\}$, where x_0 satisfies $x_0^2 \equiv a \pmod{p}$. For every $x \in A$ not equal to x_0 or $p-x_0$, we have $f(x) \neq x$. So we may pair the rest of the elements as $\{x, f(x)\}$ as before, and there are $\frac{p-3}{2}$ such pairs.

- (2) Prove, by multiplying the numbers $1, \dots, p-1$ in two ways and reducing the product modulo p , that

$$(p-1)! \equiv -\left(\frac{a}{p}\right) a^{\frac{p-1}{2}} \pmod{p}. \quad (5)$$

Solution. If $a \not\equiv 1 \pmod{p}$, we have by part (1) that

$$(p-1)! = \prod_{\{x, f(x)\} \text{ is a pair}} xf(x) \equiv a^{\frac{p-1}{2}} \pmod{p};$$

if $a \equiv 1 \pmod{p}$, part (1) implies that

$$(p-1)! = x_0(p-x_0) \prod_{\substack{\{x, f(x)\} \text{ is a pair} \\ x \neq x_0, p-x_0}} xf(x) \equiv -x_0^2 \cdot a^{\frac{p-3}{2}} \equiv -a \cdot a^{\frac{p-3}{2}} \equiv -a^{\frac{p-1}{2}} \pmod{p}.$$

The two cases can be merged into (5) using the definition

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \equiv 1 \pmod{p}, \\ -1, & \text{if } a \not\equiv 1 \pmod{p}. \end{cases}$$

- (3) Deduce Wilson's theorem that $(p-1)! \equiv -1 \pmod{p}$ and then Euler's criterion.

Solution. Wilson's theorem that $(p-1)! \equiv -1 \pmod{p}$ is an immediate consequence of (5) with $a = 1$. Substituting it into (5) gives

$$-1 \equiv -\left(\frac{a}{p}\right) a^{\frac{p-1}{2}} \pmod{p}.$$

Multiplying both sides by $-\left(\frac{a}{p}\right)$ gives Euler's criterion.

5. (Gauss's lemma) Use Gauss's lemma to show that $-3 \text{ R } p$ if $p \equiv 1 \pmod{6}$ and $-3 \text{ N } p$ if $p \equiv 5 \pmod{6}$.

Solution. The primes p in consideration are either $\equiv 1 \pmod{6}$ or $\equiv 5 \pmod{6}$. By Gauss's lemma,

$$\left(\frac{a}{p}\right) = (-1)^m (\varepsilon_{-3})^{\frac{p-1}{2}},$$

where $\varepsilon_{-3} = -1$ records the sign of -3 and

$$m \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{3k}{p} \right\rfloor + \frac{p^2-1}{8}(-3-1) \equiv \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{3k}{p} \right\rfloor \pmod{2}.$$

To evaluate the sum, note that

$$\left\lfloor \frac{3k}{p} \right\rfloor = \begin{cases} 0, & \text{if } k < p/3, \\ 1, & \text{if } p/3 < k \leq (p-1)/2. \end{cases}$$

So the sum equals the number of integers $p/3 < k \leq (p-1)/2$, which is

$$\frac{p-1}{2} - \lfloor p/3 \rfloor.$$

Hence

$$\left(\frac{a}{p}\right) = (-1)^{(p-1)-\lfloor p/3 \rfloor} = (-1)^{\lfloor p/3 \rfloor}.$$

If $p \equiv 1 \pmod{6}$, then $\frac{p-1}{3}$ is the greatest integer not exceeding $p/3$. Since this integer is even (e.g. writing $p = 6q + 1$ gives $\frac{p-1}{3} = 2q$), we have

$$\left(\frac{a}{p}\right) = (-1)^{\frac{p-1}{3}} = 1,$$

so $-3 \text{ R } p$. Similarly, if $p \equiv 5 \pmod{6}$, then $\frac{p-2}{3}$ is the greatest integer not exceeding $p/3$. Since this integer is odd (e.g. writing $p = 6q + 5$ gives $\frac{p-2}{3} = 2q + 1$), we have

$$\left(\frac{a}{p}\right) = (-1)^{\frac{p-2}{3}} = -1.$$

Thus $-3 \text{ N } p$.

6. (Quadratic reciprocity) Evaluate the following Legendre symbols.

$$(1) \left(\frac{17}{41}\right)$$

Solution. Since $17 \equiv 1 \pmod{4}$ and $41 \equiv 1 \pmod{4}$, we have by quadratic reciprocity that

$$\left(\frac{17}{41}\right) = \left(\frac{41}{17}\right) = \left(\frac{7}{17}\right) = \left(\frac{17}{7}\right) = \left(\frac{3}{7}\right).$$

Since $3 \equiv 3 \pmod{4}$ and $7 \equiv 3 \pmod{4}$, applying quadratic reciprocity again gives

$$\left(\frac{17}{41}\right) = \left(\frac{3}{7}\right) = -\left(\frac{7}{3}\right) = -\left(\frac{1}{3}\right) = -1.$$

(2) $\left(\frac{15}{43}\right)$

Solution. By complete multiplicativity, we have

$$\left(\frac{15}{43}\right) = \left(\frac{3}{43}\right) \left(\frac{5}{43}\right).$$

Since $3 \equiv 3 \pmod{4}$ and $43 \equiv 3 \pmod{4}$, we have by quadratic reciprocity that

$$\left(\frac{3}{43}\right) = -\left(\frac{43}{3}\right) = -\left(\frac{1}{3}\right) = -1.$$

Similarly, since $5 \equiv 1 \pmod{4}$, applying quadratic reciprocity again gives

$$\left(\frac{5}{43}\right) = \left(\frac{43}{5}\right) = \left(\frac{3}{5}\right) = \left(\frac{5}{3}\right) = \left(\frac{-1}{3}\right) = -1,$$

where in the last step we used $5 \equiv -1 \pmod{3}$ and $\left(\frac{-1}{3}\right) = -1$ due to $3 \equiv 3 \pmod{4}$.

Putting everything together, we get

$$\left(\frac{15}{43}\right) = \left(\frac{3}{43}\right) \left(\frac{5}{43}\right) = (-1)(-1) = 1.$$

(3) $\left(\frac{280}{293}\right)$

Solution. By complete multiplicativity, we have

$$\left(\frac{280}{293}\right) = \left(\frac{2}{293}\right)^3 \left(\frac{5}{293}\right) \left(\frac{7}{293}\right) = \left(\frac{2}{293}\right) \left(\frac{5}{293}\right) \left(\frac{7}{293}\right).$$

Since $293 \equiv 5 \pmod{8}$, we know that

$$\left(\frac{2}{293}\right) = -1.$$

Since $5 \equiv 1 \pmod{4}$, quadratic reciprocity gives

$$\left(\frac{5}{293}\right) = \left(\frac{293}{5}\right) = \left(\frac{3}{5}\right) = -1,$$

by the calculations done in part (2). Similarly, since $293 \equiv 1 \pmod{4}$, quadratic reciprocity leads to

$$\left(\frac{7}{293}\right) = \left(\frac{293}{7}\right) = \left(\frac{-1}{7}\right) = -1,$$

where in the last step we used $293 \equiv -1 \pmod{7}$ and $\left(\frac{-1}{7}\right) = -1$ due to $7 \equiv 3 \pmod{4}$.

Putting everything together, we get

$$\left(\frac{280}{293}\right) = \left(\frac{2}{293}\right) \left(\frac{5}{293}\right) \left(\frac{7}{293}\right) = (-1)(-1)(-1) = -1.$$

7. (Jacobi symbol) For any $a \in \mathbb{Z}$ and any positive odd integer $n = p_1 \cdots p_k$, where $p_1, \dots, p_k$ are odd primes, define the *Jacobi symbol*

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right) \cdots \left(\frac{a}{p_k}\right)$$

with the convention that $\left(\frac{a}{1}\right) = 1$. It is understood that each $\left(\frac{a}{p_i}\right)$ on the right-hand side is a Legendre symbol. In particular, the Jacobi symbol agrees with the Legendre symbol when n is a prime. Verify the following properties for the Jacobi symbol.

(1) (Periodicity) For any positive odd integer n and any $a, b \in \mathbb{Z}$ such that $a \equiv b \pmod{n}$,

$$\left(\frac{a}{n}\right) = \left(\frac{b}{n}\right).$$

Solution. Writing $n = p_1 \cdots p_k$ and using periodicity of the Legendre symbols, we have

$$\left(\frac{a}{n}\right) = \prod_{i=1}^k \left(\frac{a}{p_i}\right) = \prod_{i=1}^k \left(\frac{b}{p_i}\right) = \left(\frac{b}{n}\right).$$

(2) (Complete multiplicativity) For any positive odd integers m, n and any $a, b \in \mathbb{Z}$,

$$\begin{aligned} \left(\frac{ab}{n}\right) &= \left(\frac{a}{n}\right) \left(\frac{b}{n}\right), \\ \left(\frac{a}{mn}\right) &= \left(\frac{a}{m}\right) \left(\frac{a}{n}\right). \end{aligned}$$

Solution. Let $m = p_1 \cdots p_k$ and $n = q_1 \cdots q_\ell$. Using complete multiplicativity of the Legendre symbols, we have

$$\left(\frac{ab}{n}\right) = \prod_{j=1}^{\ell} \left(\frac{ab}{q_j}\right) = \prod_{j=1}^{\ell} \left(\frac{a}{q_j}\right) \left(\frac{b}{q_j}\right) = \prod_{j=1}^{\ell} \left(\frac{a}{q_j}\right) \prod_{j=1}^{\ell} \left(\frac{b}{q_j}\right) = \left(\frac{a}{n}\right) \left(\frac{b}{n}\right).$$

The second identity follows merely from the definition of Jacobi symbols:

$$\left(\frac{a}{mn}\right) = \left(\frac{a}{p_1 \cdots p_k q_1 \cdots q_\ell}\right) = \prod_{i=1}^k \left(\frac{a}{p_i}\right) \prod_{j=1}^{\ell} \left(\frac{a}{q_j}\right) = \left(\frac{a}{m}\right) \left(\frac{a}{n}\right).$$

(3) $(-1 \pmod{n})$ For any positive odd integer n ,

$$\left(\frac{-1}{n}\right) = (-1)^{\frac{n-1}{2}}.$$

Solution. Following the hint, we first prove

$$\frac{a_1 a_2 \cdots a_k - 1}{2} \equiv \sum_{i=1}^k \frac{a_i - 1}{2} \pmod{2} \quad (6)$$

for any odd $a_1, \dots, a_k \in \mathbb{Z}$. This is trivial when $k = 1$. For $k = 2$, we have

$$\frac{a_1 a_2 - 1}{2} - \frac{a_1 - 1}{2} - \frac{a_2 - 1}{2} = \frac{a_1 a_2 - a_1 - a_2 + 1}{2} = \frac{(a_1 - 1)(a_2 - 1)}{2} \equiv 0 \pmod{2},$$

since a_1, a_2 being odd implies that $a_1 - 1, a_2 - 1$ are even. Thus

$$\frac{a_1 a_2 - 1}{2} \equiv \frac{a_1 - 1}{2} + \frac{a_2 - 1}{2} \pmod{2},$$

confirming the case $k = 2$. Suppose now that (6) holds for some k and any odd $a_1, \dots, a_k \in \mathbb{Z}$. Given any odd $a_1, \dots, a_{k+1} \in \mathbb{Z}$, the case $k = 2$ and inductive hypothesis imply

$$\begin{aligned} \frac{a_1 a_2 \cdots a_{k+1} - 1}{2} &= \frac{(a_1 a_2 \cdots a_k) a_{k+1} - 1}{2} \equiv \frac{a_1 a_2 \cdots a_k - 1}{2} + \frac{a_{k+1} - 1}{2} \\ &\equiv \sum_{i=1}^k \frac{a_i - 1}{2} + \frac{a_{k+1} - 1}{2} \\ &\equiv \sum_{i=1}^{k+1} \frac{a_i - 1}{2} \pmod{2}. \end{aligned}$$

Hence (6) holds for $k + 1$ and any odd $a_1, \dots, a_{k+1} \in \mathbb{Z}$. By induction, (6) holds for all k and any odd $a_1, \dots, a_k \in \mathbb{Z}$.

Let $n = p_1 \cdots p_k$. Then $\left(\frac{-1}{p_i}\right) = (-1)^{\frac{p_i-1}{2}}$ for every $1 \leq i \leq k$. Apply (6) with $a_i = p_i$, we obtain

$$\left(\frac{-1}{n}\right) = \prod_{i=1}^k \left(\frac{-1}{p_i}\right) = \prod_{i=1}^k (-1)^{\frac{p_i-1}{2}} = (-1)^{\sum_{i=1}^k \frac{p_i-1}{2}} = (-1)^{\frac{p_1 \cdots p_k - 1}{2}} = (-1)^{\frac{n-1}{2}}.$$

(4) (2 mod n) For any positive odd integer n ,

$$\left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}}.$$

Solution. Following the hint, we first show

$$\frac{(a_1 a_2 \cdots a_k)^2 - 1}{8} \equiv \sum_{i=1}^k \frac{a_i^2 - 1}{8} \pmod{2} \quad (7)$$

for any odd $a_1, \dots, a_k \in \mathbb{Z}$. This is trivial when $k = 1$. For $k = 2$, we have

$$\frac{(a_1 a_2)^2 - 1}{8} - \frac{a_1^2 - 1}{8} - \frac{a_2^2 - 1}{8} = \frac{a_1^2 a_2^2 - a_1^2 - a_2^2 + 1}{8} = \frac{(a_1^2 - 1)(a_2^2 - 1)}{8} \equiv 0 \pmod{2},$$

since a_1, a_2 being odd implies $a_1^2 \equiv 1 \pmod{8}$ and $a_2^2 \equiv 1 \pmod{8}$. Thus

$$\frac{(a_1 a_2)^2 - 1}{8} \equiv \frac{a_1^2 - 1}{8} + \frac{a_2^2 - 1}{8} \pmod{2},$$

confirming the case $k = 2$. Suppose now that (7) holds for some k and any odd $a_1, \dots, a_k \in \mathbb{Z}$. Given any odd $a_1, \dots, a_{k+1} \in \mathbb{Z}$, the case $k = 2$ and inductive hypothesis imply

$$\begin{aligned} \frac{(a_1 a_2 \cdots a_{k+1})^2 - 1}{8} &= \frac{[(a_1 a_2 \cdots a_k) a_{k+1}]^2 - 1}{8} \equiv \frac{(a_1 a_2 \cdots a_k)^2 - 1}{8} + \frac{a_{k+1}^2 - 1}{8} \\ &\equiv \sum_{i=1}^k \frac{a_i^2 - 1}{8} + \frac{a_{k+1}^2 - 1}{8} \\ &\equiv \sum_{i=1}^{k+1} \frac{a_i^2 - 1}{8} \pmod{2}. \end{aligned}$$

Hence (7) holds for $k + 1$ and any odd $a_1, \dots, a_{k+1} \in \mathbb{Z}$. By induction, (7) holds for all k and any odd $a_1, \dots, a_k \in \mathbb{Z}$.

Let $n = p_1 \cdots p_k$. Then $\left(\frac{2}{p_i}\right) = (-1)^{\frac{p_i^2-1}{8}}$ for every $1 \leq i \leq k$. Apply (7) with $a_i = p_i$, we obtain

$$\left(\frac{2}{n}\right) = \prod_{i=1}^k \left(\frac{2}{p_i}\right) = \prod_{i=1}^k (-1)^{\frac{p_i^2-1}{8}} = (-1)^{\sum_{i=1}^k \frac{p_i^2-1}{8}} = (-1)^{\frac{(p_1 \cdots p_k)^2 - 1}{8}} = (-1)^{\frac{n^2-1}{8}}.$$

(5) (Quadratic reciprocity) For any positive odd integers m, n with $\gcd(m, n) = 1$,

$$\left(\frac{m}{n}\right) \left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}} = \begin{cases} -1, & \text{if } m, n \equiv 3 \pmod{4}, \\ 1, & \text{otherwise.} \end{cases}$$

Solution. Let $m = p_1 \cdots p_k$ and $n = q_1 \cdots q_\ell$. The law of quadratic reciprocity for Legendre symbols gives

$$\left(\frac{p_i}{q_j}\right) \left(\frac{q_j}{p_i}\right) = (-1)^{\frac{p_i-1}{2} \cdot \frac{q_j-1}{2}}$$

for $1 \leq i \leq k$ and $1 \leq j \leq \ell$. Since

$$\left(\frac{m}{n}\right) = \prod_{i=1}^k \prod_{j=1}^{\ell} \left(\frac{p_i}{q_j}\right),$$

$$\left(\frac{n}{m}\right) = \prod_{i=1}^k \prod_{j=1}^{\ell} \left(\frac{q_j}{p_i}\right),$$

we have

$$\left(\frac{m}{n}\right) \left(\frac{n}{m}\right) = \prod_{i=1}^k \prod_{j=1}^{\ell} \left(\frac{p_i}{q_j}\right) \left(\frac{q_j}{p_i}\right) = \prod_{i=1}^k \prod_{j=1}^{\ell} (-1)^{\frac{p_i-1}{2} \cdot \frac{q_j-1}{2}} = (-1)^{\sum_{i=1}^k \sum_{j=1}^{\ell} \frac{p_i-1}{2} \cdot \frac{q_j-1}{2}}.$$

By (6), we see that

$$\begin{aligned} \sum_{i=1}^k \frac{p_i - 1}{2} &\equiv \frac{p_1 p_2 \cdots p_k - 1}{2} \equiv \frac{m - 1}{2} \pmod{2}, \\ \sum_{j=1}^{\ell} \frac{q_j - 1}{2} &\equiv \frac{q_1 q_2 \cdots q_\ell - 1}{2} \equiv \frac{n - 1}{2} \pmod{2}. \end{aligned}$$

Hence

$$\sum_{i=1}^k \sum_{j=1}^{\ell} \frac{p_i - 1}{2} \cdot \frac{q_j - 1}{2} = \sum_{i=1}^k \frac{p_i - 1}{2} \cdot \sum_{j=1}^{\ell} \frac{q_j - 1}{2} \equiv \frac{m-1}{2} \cdot \frac{n-1}{2} \pmod{2}.$$

It follows that

$$\left(\frac{m}{n}\right) \left(\frac{n}{m}\right) = (-1)^{\sum_{i=1}^k \sum_{j=1}^{\ell} \frac{p_i-1}{2} \cdot \frac{q_j-1}{2}} = (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}}.$$

Thus the Jacobi symbol retains all the key properties of the Legendre symbol. These properties often make evaluation of Legendre symbols a lot easier. For instance,

$$\left(\frac{15}{43}\right) = -\left(\frac{43}{15}\right) = -\left(\frac{43}{5}\right) \left(\frac{43}{3}\right) = -\left(\frac{3}{5}\right) \left(\frac{1}{3}\right) = -\left(\frac{5}{3}\right) = -\left(\frac{2}{3}\right) = 1.$$

8. (Sum of two squares) Determine which of the following integers can be expressed as a sum of two squares. For each one that can, find such a representation.

(1) 605

Solution. The prime factorization $605 = 5 \cdot 11^2$, where $5 \equiv 1 \pmod{4}$, tells us that 605 can be expressed as a sum of two squares. Since $5 = 1^2 + 2^2$, we have

$$605 = (1^2 + 2^2) \cdot 11^2 = (1 \cdot 11)^2 + (2 \cdot 11)^2 = 11^2 + 22^2.$$

(2) 2026

Solution. Note that $2026 = 2 \cdot 1013$, where $1013 \equiv 1 \pmod{4}$ is prime. So 2026 can be expressed as a sum of two squares. Suppose $1013 = x^2 + y^2$ with $x \geq y \geq 1$. Since

$$2 \cdot 22^2 = 968 < 1013 < 32^2 = 1024,$$

we have

$$22 < \sqrt{\frac{1013}{2}} \leq x \leq \sqrt{1013} < 32.$$

So

$$x \in \{23, 24, 25, 26, 27, 28, 29, 30, 31\}.$$

If we try $x = 23$, then we get

$$y^2 = 1013 - x^2 = 1013 - 23^2 = 1013 - 529 = 484 = 18^2,$$

so $y = 18$. This gives $1013 = 23^2 + 18^2$. Multiplying this by $2 = 1^2 + 1^2$ using the formula

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$$

gives us

$$2026 = (1^2 + 1^2)(23^2 + 18^2) = (1 \cdot 23 - 1 \cdot 18)^2 + (1 \cdot 23 + 1 \cdot 18)^2 = 5^2 + 41^2.$$

(3) 6174

Solution. Since $6174 = 2 \cdot 3^2 \cdot 7^3$, we see that $7 \equiv 3 \pmod{4}$ but $e_7(6174) = 3$ is odd. So 6174 can not be expressed as a sum of two squares.

9. (Quadratic congruence equations) Solve the quadratic congruence $x^2 + x + 3 \equiv 0 \pmod{15}$.

Solution. Since $\gcd(4, 15) = 1$, we have

$$x^2 + x + 3 \equiv 0 \pmod{15} \iff 4(x^2 + x + 3) \equiv 0 \pmod{15}.$$

(It works fine if you also multiply the modulus by 4 as we did in class for the most general case, though it may involve more calculations.) Completing the square gives

$$(2x + 1)^2 + 11 \equiv 0 \pmod{15}.$$

Let $y = 2x + 1$. Then $y^2 \equiv -11 \equiv 4 \pmod{15}$. Since $2^2 = (-2)^2 = 4$, we know $y \equiv \pm 2 \pmod{15}$ are solutions. After trying the rest of the possibilities $y \equiv \pm 1, \pm 3, \pm 4, \pm 5, \pm 6, \pm 7$, we find that none of these is a solution.

To find x , we need to solve $2x + 1 \equiv 2 \pmod{15}$ and $2x + 1 \equiv -2 \pmod{15}$. The first one is equivalent to $2x \equiv 1 \pmod{15}$. It is easy to find its solution $x \equiv 8 \pmod{15}$. By definition, this is actually a multiplicative inverse of 2 $\pmod{15}$.

The second one is equivalent to $2x \equiv -3 \pmod{15}$. Multiplying both sides by 8, a multiplicative inverse of 2 $\pmod{15}$, we get $x \equiv 16x \equiv -24 \equiv 6 \pmod{15}$.

So the solutions to the original congruence are $x \equiv 6, 8 \pmod{15}$.

10. (Sum of three squares) Show that no integer of the form $4^a(8b + 7)$, where $a, b \geq 0$ are integers, can be expressed as a sum of three squares.

Solution. Let $n = 4^a(8b + 7)$. First consider the case $a = 0$, so that $n = 8b + 7 \equiv 7 \pmod{8}$. Since every square is congruent to 0, 1, 4 $\pmod{8}$, the sum of three arbitrary squares is never congruent to 7 $\pmod{8}$. Hence n is not a sum of three squares.

Now consider the case $a \geq 1$. Suppose that n is the least such integer that can be expressed as a sum of three squares:

$$n = x^2 + y^2 + z^2. \tag{8}$$

Since $a \geq 1$, we have $n \equiv 0 \pmod{4}$. Because the square of an even number is congruent to 0 $\pmod{4}$ and the square of an odd number is congruent to 1 $\pmod{4}$, (8) can hold only when x, y, z are all even. Dividing both sides of (8) by 4 yields

$$\frac{n}{4} = \left(\frac{x}{2}\right)^2 + \left(\frac{y}{2}\right)^2 + \left(\frac{z}{2}\right)^2,$$

which is a representation of $n/4 = 4^{a-1}(8b + 7)$ as a sum of three squares. This is impossible when $a = 1$, as shown above. It is also impossible when $a > 1$ because of $n/4 = 4^{a-1}(8b + 7) < n$, which would contradict the minimality hypothesis on n . Therefore, no such n exists.

We have thus confirmed that integers of the form $n = 4^a(8b + 7)$ can never be expressed as a sum of three squares.

Optional challenge

11. (Half factorials mod p) Let $p > 2$ be prime.

(1) Prove

$$\left[\left(\frac{p-1}{2} \right)! \right]^2 \equiv (-1)^{\frac{p+1}{2}} \pmod{p}.$$

Solution. The numbers $1, \dots, p-1$ can be grouped as $\{k, p-k\}$ for $1 \leq k \leq \frac{p-1}{2}$. Multiplying them out and noting that $k(p-k) \equiv -k^2 \pmod{p}$, we obtain

$$(p-1)! = \prod_{k=1}^{\frac{p-1}{2}} k(p-k) \equiv \prod_{k=1}^{\frac{p-1}{2}} (-k^2) \equiv (-1)^{\frac{p-1}{2}} \prod_{k=1}^{\frac{p-1}{2}} k^2 \equiv (-1)^{\frac{p-1}{2}} \left[\left(\frac{p-1}{2} \right)! \right]^2 \pmod{p}.$$

By Wilson's theorem, $(p-1)! \equiv -1 \pmod{p}$. Thus

$$-1 \equiv (-1)^{\frac{p-1}{2}} \left[\left(\frac{p-1}{2} \right)! \right]^2 \pmod{p}.$$

Multiplying both sides by $(-1)^{\frac{p-1}{2}}$ proves the asserted congruence.

(2) Show that if $p \equiv 3 \pmod{4}$, then

$$\left(\frac{p-1}{2} \right)! \equiv (-1)^n \pmod{p},$$

where n is the number of integers $1 \leq k < p/2$ that are quadratic nonresidues modulo p .

Solution. Since $p \equiv 3 \pmod{4}$, part (1) tells us that

$$\left[\left(\frac{p-1}{2} \right)! \right]^2 \equiv 1 \pmod{p},$$

which can be rewritten as

$$\left[\left(\frac{p-1}{2} \right)! + 1 \right] \left[\left(\frac{p-1}{2} \right)! - 1 \right] \equiv 0 \pmod{p}.$$

By Euclid's lemma, we have

$$\prod_{k=1}^{\frac{p-1}{2}} k = \left(\frac{p-1}{2} \right)! \equiv \varepsilon \pmod{p}, \tag{9}$$

where $\varepsilon = \pm 1$. To determine the exact value of ε , we take the Legendre symbols $\pmod{p}$ of both sides and use periodicity and complete multiplicativity:

$$\prod_{k=1}^{\frac{p-1}{2}} \left(\frac{k}{p} \right) = \left(\frac{\varepsilon}{p} \right). \tag{10}$$

Since $p \equiv 3 \pmod{4}$ implies $\left(\frac{-1}{p} \right) = -1$, we have

$$\begin{aligned} \left(\frac{\varepsilon}{p} \right) &= \begin{cases} 1, & \text{if } \varepsilon = 1, \\ -1, & \text{if } \varepsilon = -1, \end{cases} \\ &= \varepsilon. \end{aligned} \tag{11}$$

In addition, it follows from the definition

$$\left(\frac{k}{p} \right) = \begin{cases} 1, & \text{if } k \text{ R } p, \\ -1, & \text{if } k \text{ N } p, \end{cases}$$

that

$$\prod_{k=1}^{\frac{p-1}{2}} \left(\frac{k}{p} \right) = (-1)^n,$$

where n is the number of $1 \leq k < p/2$ such that $k \nmid p$. Plugging this and (11) into (10) gives $\varepsilon = (-1)^n$. Finally, combining this with (9) proves the required congruence.